MPJ Assignment :- 02

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PROBLEM STATEMENT :- Write a program in Java showing hierarchial inheritance with base class as Employee & derived class as Full time employee & Intern Employee with methods Displaysalary in base class and calculatesalary in derived classes. Calculate the Salary method will calculate as per increment given to fulltime & intern employees. Fulltime - 50% hike, Intern - 25% Display salary before & after hike.

THEORY :-

1] What is inheritance in Java?

⇒ It is a mechanism where one class acquires the properties & methods of another class.
The class that is inherited from is called the parent class. or superclass & the class that inherits is called the child or subclass.
It helps in reusing existing code and establishing a relationship between classes.

2] Why do we use inheritance?

⇒ i] It promotes code reusability.
ii] It helps in better code organization.
iii] It supports method overriding which helps in achieving runtime polymorphism.
iv] It makes programs easy to maintain & extend.

3] Three types of inheritance?

⇒ 1] Single Inheritance

One child class inherits from one parent class.

2] Multilevel Inheritance

A class inherits from another class which itself is inherited from another class.

3] Hierarchial Inheritance

Multiple child classes inherit from single parent class.

FAQs :-

1] Is multiple inheritance supported in Java? How is it achieved?

⇒ Java does not support multiple inheritance using classes because it can cause ambiguity.
But, it does support multiple inheritance using interfaces.
It is achieved by implementing more than one interface in a class.

2] What is a relationship in Java?

⇒ An Is-A relationship represents inheritance in Java. It means one class is a type of another class.
Eg- If Dog extends Animal, then Dog is an Animal.
This shows Is-A relationship & is implemented using extend keyword. This helps in reuse of code and supports polymorphism.

3] Are constructor & instance initialization block inherited to subclass?

→ No, the constructor & instance intialization blocks are not inherited by subclass.
However, the constructor of the parent class is called when a subclass object is created using super().
The instance initialization block of the parent class executes but it still is not inherited. So, they are executed during object creation but inheritance does not apply to them.